

COS10005 – Web Development - Assignment 2

Swinburne University of Technology School of Science, Computing and Engineering Technologies

Web Development COS10005

Assignment 2, SEMESTER 1, 2026

Semester 1, 2026

Due Date: 31 May 2026, 11:59 PM (Melbourne Time)

Weighting: 40%

Purpose of the Assignment

This is an **individual assignment**. The goal is to apply your knowledge of **HTML5** and **CSS3** to design, develop, and test a small but complete website for a real-world scenario. Meanwhile, JavaScript will be used to validate user input data before it is submitted to the server for processing. Moreover, JavaScript will be used to add more interactivity to the webpages.

Requirements and Specifications

The assignment requires that you develop a website for a **restaurant discovery and reservation platform**, which will allow users to:

- Learn about the platform
- Browse available restaurants on your platform
- Receive restaurant recommendations based on user preferences
- Register for an account (This page is only a front-end form with JavaScript validation. No databases are required.)
- Make restaurant reservations with a deposit option.

The project will be hosted and tested on the **Mercury server**.

Task 1: Website Design

1.1 Required File Structure

Students must follow the folder structure below exactly:

```
assignment2/
├─ index.html                # Home page
├─ restaurants.html          # Restaurant listing page
├─ recommend.html           # Restaurant recommendation page
├─ register.html            # User registration page
├─ reservation.html         # Reservation page
├─ bill.html                # Optional, details refer to Bonus Task
Section
├─ XXXXXXXX.html            # Any other HTML pages based on your
design
├─ css/
│   └─ style.css
├─ js/
│   └─ script.js
├─ images/
│   └─ logo.png
│   └─ restaurant1.jpg ...
└─ Readme.txt               # Details refer to Submission Requirements
```

1.2 Site Structure and Layout

- A consistent navigation menu must be present on all pages.
- Users must be able to navigate to all other pages from any page.
- Navigation highlighting: current page highlighted in the nav bar
- HTML5 semantic elements must be used, including:
 - `<header>`, `<nav>`, `<section>`, `<footer>`
- Additional structural elements such as `<div>` may be used where appropriate.

Task 2: Website Development

2.1 HTML Pages

All HTML files must comply with HTML5 standards. Deprecated tags and attributes are not permitted.

(a) index.html – Home Page

- Introduce the platform, for example, the purpose and values of the platform (e.g. convenience, variety, accessibility, dining experience), the target users (e.g. families,

couples, professionals, tourists), the types of services provided (restaurant discovery, recommendations, reservations)

- Include at least **one relevant image** (with reference if not original)

(b) restaurants.html – Restaurant Listing Page

This page must include at least six restaurants.

Each restaurant must include:

- Restaurant name
- Cuisine type
- Signature dishes with prices
- Deposit Amount for Reservation
- Short description (50–100 words)
- At least one image
- Average price or price range per person

(c) recommend.html – Recommendation Page

This page must provide a recommendation form allowing users to select:

- Dietary preference (e.g., Vegan, Halal, None)
- Budget range
- Dining purpose (e.g., Family, Date, Business)

To display recommended restaurants, **JavaScript** must be used to evaluate the user's selections and apply **meaningful rule-based logic** to match suitable restaurants.

The recommended restaurant should be selectable and passed to the reservation page.

(d) register.html – Registration Page

The registration page must include a form with the following fields:

- Username
- Email Address
- Phone Number
- Password and Confirm Password
- Gender
- Dietary Preferences
- Country/Region
- Submit

The form must use JavaScript validation.

(e) reservation.html – Reservation Page

The reservation page must contain a form including:

- Full name
- Email address
- Phone number
- Restaurant selection (dropdown menu, pre-filled if coming from recommendation page or home page)
- Reservation date and time (must not be in the past)
- Number of people (must be greater than 0)
- Special requests (optional)
- Deposit amount (predefined for each restaurant and dynamically updated based on the selected restaurant)
- Deposit method (Voucher / Online payment)
- Voucher Code (textbox, 12 digits, show only if Voucher is selected as the payment method)
- Credit card details (textbox, shown only if online payment selected; Visa/MC=16 digits, Amex=15 digits). **Note:** Please use *fake card information* for testing. No real validation is required.
- Billing email address (with a “Same as email address” option that automatically fills this field with the previously entered email address when selected)

Users can access this page either by selecting a restaurant from the home page or via the recommendation page.

If users enter this page directly, they can choose a restaurant from the Restaurant selection dropdown menu.

*** Users can always change the Restaurant selection at any time on the reservation page.**

* Form method must be set to **POST**, and the action must point to:

<http://mercury.swin.edu.au/it000000/cos10005/regtest.php>

2.2 CSS Requirements

- At least **one external CSS file** (**style.css**) must be linked to all pages.
- **Responsive design** is required:
 - Desktop: $\geq 1024\text{px}$
 - Tablet: $768\text{--}1023\text{px}$
 - Mobile: $\leq 767\text{px}$
- **Accessibility requirements:**
 - All images must include appropriate **alt** text
 - Sufficient colour contrast
 - Readable font sizes
 - Logical heading hierarchy

2.3 JavaScript Requirements

All JavaScript code must be contained in `script.js`.

(a) Registration Form Validation

- All fields must be completed
- Username: minimum 5 characters; letters, numbers, and underscores only
- Email: valid email format
- Phone number: digits only, 8–15 digits
- Password: minimum 10 characters, including uppercase, lowercase, numbers, and special characters
- Confirm password must match the password
- Gender must be selected

Clear error messages must be displayed, and form submission must be blocked until all errors are resolved.

(b) Reservation Form Validation

- Required fields must not be empty
- Email must be valid (in valid format)
- Phone number must contain at least 10 digits
- Reservation date must not be in the past
- The number of people must be greater than 0
- Payment method:
 - If Voucher → no credit card fields are required or displayed; instead, a textbox for a 12-digit voucher code is shown, no need to validate.
 - If Online payment → Credit card number required (digits only).
 - Visa/Mastercard: 16 digits
 - Amex: 15 digits
 - Format check only; no real processing.

(c) Error Handling

- Errors must be displayed to users clearly
- Form submission must be blocked until all errors are fixed

Bonus Task (Optional, +5 Marks)

An Extra **Estimated Bill Calculator Page**(Optional)

Students may implement an estimated bill calculator that:

- Calculates an estimated total cost (for example, based on group size and the restaurant's price range, or optionally selected dishes)
- Is designed fully by you (you may refer to real-world scenarios for inspiration)
- Updates dynamically using JavaScript

This bonus:

- Is optional
- Is not required to achieve full marks (100/100)
- Can earn up to +5 bonus marks, capped at 100 overall

Individual Interview

In Week 12, each student must attend a 5-minute individual interview with the lecturer or tutor. This interview is worth **15%** of the total assignment mark.

Interview Structure:

- 2–3 minutes: Student demonstrates their website and explains key features and design decisions
- 2–3 minutes: Lecturer asks targeted questions about the student's own code and logic

Academic Integrity Notice:

If a student is unable to reasonably explain their own work, demonstrate understanding of their code, or justify key design decisions, the submission may be considered not genuinely produced by the student.

This may result in:

- Significant mark deductions for the assignment, or even
- A failed result for the entire assignment.

Submission Requirements

Before submitting the assignment, please ensure that you have completed or considered the following:

- Check Canvas for announcements related to the assignment.
- All HTML and CSS must pass **W3C validation**
- Check your assignment against the attached marking rubric
- All files must run correctly on the **Mercury server**. If your assignment cannot be displayed on Mercury properly, penalties will apply.
- **In addition to submission via Canvas**, the project should also be submitted to **GitHub** as a backup, and the repository must be accessible to the marker during the marking

period. Failure to comply will result in mark deductions. Please include the **GitHub repository link** in the Readme.txt. file

- The assignment is your original work. If your assignment is not your original work, your result will be 0 marks for this assignment.
- Provide references for all third-party content (including the images) used on your webpages in the Readme.txt.
- Submission must include:
 - All HTML, CSS, JS, and images
 - **Readme.txt** describing:
 - Website structure
 - GitHub repository link
 - Explanation of JavaScript validation logic (in plain English)
 - Any known issues or limitations
 - Any references, if applied
- For Canvas submission, files must be compressed into: wd_assign2_studentID.zip

Marking Rubric

Criteria	Pts
All required HTML pages are correctly created and make appropriate use of HTML5 semantic elements.	5
The webpage layout and visual consistency are coherent and appropriate for the platform, and the overall UI design is intuitive and visually appealing.	8
The home page is designed as required (clearly introducing the platform, its purpose, target users, services, and so on).	3
At least six (6) restaurants are displayed, each including all required details.	3
The registration page includes all required form fields and controls.	4
The reservation page includes all required fields, deposit logic, and restaurant selection behaviour	4
HTML code is properly indented, organised, commented and fully validated (1 mark deducted per error).	5
The external CSS file is correctly linked and applied to all pages.	2
Responsive design is implemented for desktop, tablet, and mobile.	3
Accessibility features are implemented (Including but not limited to alt text, sufficient colour contrast, readable font sizes, and logical heading hierarchy).	3
CSS code quality (CSS code is organised, commented, and validated, 1 mark deducted per error).	5

Navigation is clear and consistent, and current page highlighting works correctly.	3
All required registration validations are implemented with clear error messages.	6
Reservation form validation is correctly implemented, including deposit and payment logic.	5
Meaningful rule-based recommendation logic that clearly differentiates outcomes based on different user inputs.	4
Conditional field display and dynamic updates function correctly.	4
JavaScript code quality (including but not limited to well-organised, commented, and readable).	6
The JS file is correctly linked and applied to all pages.	2
The Readme.txt file is constructed properly as required.	4
Professionalism & Submission (Including: correct folder structure, successful deployment on the Mercury server, and a GitHub repository created as required.)	6
Interview: Website demonstration, students could clearly demonstrate their website and the implemented features.	5
Interview: Explanation of code & logic, students could explain their JavaScript logic, data structures, and key design decisions.	5
Interview: Response to questions, students could accurately answer targeted questions about their own code.	5
Total	100
Bonus: An Extra Estimated Bill Calculator Page is created, and the overall design and implementation quality is outstanding	5